

VILLYAN SUTANTO

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SUMMARY

Computer Science undergraduate at BINUS University with a strong mathematical foundation and hands-on experience building end-to-end systems across machine learning, IoT, spatial analysis, and optimization. Developed projects including YOLOv8-based traffic signal optimization, an RFID and ultrasonic sensor-integrated IoT door lock, and a PostGIS-backed WebGIS platform. Seeking internship opportunities in Data Analysis, AI Engineering, or GIS Engineering.

EDUCATION

Bina Nusantara (BINUS) University - <i>School of Computer Science</i>	2023 - 2027
Bachelor's degree in Computer Science and Mathematics	Jakarta, Indonesia
- GPA: 3.85/4.00 - Relevant coursework: Computational Geometry, Abstract Algebra, Real Analysis, Mathematical Modelling, Cloud Architecture (AWS), Data Structures, Algorithm Design, Software Engineering.	

EXPERIENCE

Core Governing Board - <i>Regional Secretary</i>	February 2025 - 2026
Keluarga Mahasiswa Katolik (KMK) BINUS University @Kemanggisian	Jakarta, Indonesia
- Managed the flow and administration of incoming and outgoing organizational documents, including proposals, accountability reports, Memoranda of Agreement (MoA), Memoranda of Understanding (MoU), Terms of Reference (ToR), and Minutes of Meeting (MoM). - Monitored and evaluated the performance of division heads and their respective teams in executing departmental responsibilities and organizational events.	

Freshmen Mentor - <i>Freshmen Leader</i>	August 2024
First Year Program B28 BINUS University @Kemanggisian	Jakarta, Indonesia
- Guided and supported new university students in transitioning to campus life by introducing the university environment, communities, activities, academic programs, and available resources. - Assisted freshmen in adapting to university life and encouraged active participation in academic and community activities.	

Freshmen Mentor - <i>Freshmen Partner</i>	September 2024 - February 2025
First Year Program B28 BINUS University @Kemanggisian	Jakarta, Indonesia
- Guided freshmen in gaining a deeper understanding of BINUSian values, including the BINUS Graduate Attributes, Sustainable Development Goals (SDGs), and other institutional values. - Assisted freshmen in conducting a community service activity, "<i>Clean and Affordable Energy Education: Raising School Students' Awareness of Sustainable Energy Use.</i>"	

PROJECTS

StaSIUN - <i>Station Spatial Intelligence for Urban Network</i>	2026 - Present
MAPID WebGIS Competition 2026 Top 50	
Fast API · Next.js · PostGIS · Docker · QGIS · OCR · NLP	
- Building a geospatial economic intelligence platform that scores Indonesian train stations (Jakarta MRT, LRT, KRL, BRT data in QGIS) with a five-variable Spatial Economic Potential Index (SEPI). - Set up a FastAPI + Next.js monorepo on a Dockerised PostGIS database; wrote spatial queries with ST_DWithin, ST_VoronoiPolygons and GiST indexes in UTM 48S (EPSG:32748). - Audited MAPID datasets (StrukGo, MenuGo, PropertiGo, Community Maps) to design the schema; found receipt data lacked structured amounts, making an OCR pipeline a hard requirement. - Designed an S-MCDA pricing model for station naming rights: AHP + SAW scoring, TOPSIS ranking, IndoBERT/NER buyer matching and ILP portfolio optimization.	

Traffic Signal Optimization with Computer Vision and Metaheuristics

May - June 2026

Applied Mathematics Modelling and Undergraduate Thesis Research

Python · YOLOv8 · Jupyter · Queueing Simulation · Harris Hawks Optimization · OpenCV

- Integrated YOLOv8 vehicle detection with Markov queueing simulation and empirical arrival distributions to optimize signal phase timings.
- Processed ~2.9M detection records, converted counts to PCU per PKJI 2023, and calibrated saturation flow dynamically against a $\rho = 0.80$ target.
- Replaced a Genetic Algorithm with Harris Hawks Optimization.

Gojek App Review Sentiment Analysis

May - June 2026

Text Mining NLP Course Project

IndoBERT · LDA Topic Modeling · Python · Jupyter

- Classified sentiment in Indonesian-language Gojek reviews based on application versions with IndoBERT and surfaced complaint themes with LDA; co-authored an IEEE-format report.

Integrated IoT Door Lock with RFID, Ultrasonic Sensor, and Camera

May - June 2025

Embedded Systems and Internet of Things Course Project

ESP32 · Arduino IDE · C++ · Docker

- Developed an integrated smart door-lock prototype using an ESP32, combining RFID authentication, ultrasonic sensing, and camera-based monitoring.
- Implemented embedded control logic in C++ using Arduino IDE to manage sensor inputs and door-lock operations.
- Integrated multiple hardware components to support access authentication, proximity detection, and visual monitoring.
- Containerized supporting software components using Docker for a more reproducible development environment.

Keluarga Kudus Church 56th Anniversary Website

June - July 2026

Volunteer

HTML · CSS · JavaScript · Vercel

- Built and deployed the celebration website for Gereja Keluarga Kudus, Paroki Rawamangun, on Vercel with a custom domain.

SKILLS AND OTHER ACHIEVEMENTS

- Languages: Python, C++, SQL, PHP, JavaScript, TypeScript, HTML/CSS
- ML, NLP, Computer Vision: Tensorflow, Keras, PyTorch, YOLOv8, BERT, LDA, Word2Vec, RAG pipelines
- Geospatial: PostGIS, QGIS, Leaflet.js, spatial clustering, Voronoi / Delaunay methods
- Optimization & Decisions: AHP, SAW, TOPSIS, ILP, dynamic programming, Markov Chains, metaheuristics (Harris Hawks Optimization, GA)
- Engineering: FastAPI, Next.js, Docker, Git / Github, pytest (TDD), Vercel, AWS (EC2, S3, Lambda)
- Certificate: Fundamentals of Deep Learning (NVIDIA)
- Competition: TOP 50 MAPID WebGIS Competition